

**SAFETY DATA SHEET**

Creation Date 15-Dec-2011

Revision Date 25-Mar-2024

Revision Number 5

**1. Identification**

**Product Name** 4-Chlorostyrene

**Cat No. :** S20512

**CAS No** 1073-67-2  
**Synonyms** 1-Chloro-4-ethenylbenzene.; p-Chlorostyrene

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

**2. Hazard(s) identification****Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                   |            |
|-----------------------------------|------------|
| Flammable liquids                 | Category 4 |
| Acute oral toxicity               | Category 4 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 2 |

**Label Elements****Signal Word**

Warning

**Hazard Statements**

Combustible liquid  
Harmful if swallowed  
Causes skin irritation  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing and wash before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects

**Other hazards**

Contains a known or suspected endocrine disruptor.

### 3. Composition/Information on Ingredients

| Component                    | CAS No    | Weight % |
|------------------------------|-----------|----------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | >95      |
| 4-tert-Butyl catechol        | 98-29-3   | 0.1      |

### 4. First-aid measures

|                       |                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b> | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>     | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>      | Clean mouth with water and drink afterwards plenty of water.                                                      |

|                                            |                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Most important symptoms and effects</b> | . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                 |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Water mist may be used to cool closed containers. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                         |
| <b>Flash Point</b>                      | 64 °C / 147.2 °F                                                                                                                                                                 |
| <b>Method -</b>                         | No information available                                                                                                                                                         |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                         |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                  |
| <b>Upper</b>                            | No data available                                                                                                                                                                |
| <b>Lower</b>                            | No data available                                                                                                                                                                |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                         |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                         |

### Specific Hazards Arising from the Chemical

Combustible material. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.                                                                          |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.                                                                                                                    |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition.                               |
| <b>Storage.</b> | Keep away from heat, sparks and flame. Protect from direct sunlight. Keep containers tightly closed in a dry, cool and well-ventilated place. To maintain product quality: Keep refrigerated. Incompatible Materials. Acids. Bases. Halogens. Reducing Agent. |

## 8. Exposure controls / personal protection

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure |
|----------------------------|----------------------------------------------------------------------------------|

limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                              |
|----------------------------------------|------------------------------|
| Physical State                         | Liquid                       |
| Appearance                             | Clear                        |
| Odor                                   | Strong                       |
| Odor Threshold                         | No information available     |
| pH                                     | No information available     |
| Melting Point/Range                    | -16 °C / 3.2 °F              |
| Boiling Point/Range                    | 192 °C / 377.6 °F @ 760 mmHg |
| Flash Point                            | 64 °C / 147.2 °F             |
| Evaporation Rate                       | No information available     |
| Flammability (solid,gas)               | Not applicable               |
| Flammability or explosive limits       |                              |
| Upper                                  | No data available            |
| Lower                                  | No data available            |
| Vapor Pressure                         | 0.816 mmHg @ 25 °C           |
| Vapor Density                          | 4.8 (Air = 1.0)              |
| Specific Gravity                       | 1.16                         |
| Solubility                             | Insoluble in water           |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | No information available     |
| Decomposition Temperature              | No information available     |
| Viscosity                              | No information available     |
| Molecular Formula                      | C8 H7 Cl                     |
| Molecular Weight                       | 138.6                        |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

**Stability**

Stable under normal conditions.

**Conditions to Avoid**

Keep away from open flames, hot surfaces and sources of ignition. Excess heat. Exposure to light. Incompatible products.

**Incompatible Materials**

Acids, Bases, Halogens, Reducing Agent

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas

**Hazardous Polymerization**

Hazardous polymerization may occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

### Product Information

#### Component Information

| Component                    | LD50 Oral          | LD50 Dermal        | LC50 Inhalation |
|------------------------------|--------------------|--------------------|-----------------|
| Benzene, 1-chloro-4-ethenyl- | 5200 mg/kg ( Rat ) | Not listed         | Not listed      |
| 4-tert-Butyl catechol        | 815 mg/kg ( Rat )  | 1331 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic** No information available

### Products

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes and skin

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                    | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------------|-----------|------------|------------|------------|------------|------------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | Not listed | Not listed | Not listed | Not listed | Not listed |
| 4-tert-Butyl catechol        | 98-29-3   | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                    | Freshwater Algae | Freshwater Fish      | Microtox                                                                     | Water Flea         |
|------------------------------|------------------|----------------------|------------------------------------------------------------------------------|--------------------|
| Benzene, 1-chloro-4-ethenyl- | Not listed       | Not listed           | EC50 = 1.13 mg/L 5 min<br>EC50 = 1.39 mg/L 15 min<br>EC50 = 1.67 mg/L 30 min | Not listed         |
| 4-tert-Butyl catechol        | Not listed       | LC50 = 0.12 mg/L 96h | Not listed                                                                   | EC50=0.48 mg/L 48h |

**Persistence and Degradability** Insoluble in water May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component                    | log Pow |
|------------------------------|---------|
| Benzene, 1-chloro-4-ethenyl- | 3.54    |
| 4-tert-Butyl catechol        | 1.98    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3082  
 Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
 Technical Name 4-tert-Butyl catechol  
 Hazard Class 9  
 Packing Group III

#### TDG

UN-No UN3082  
 Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
 Hazard Class 9  
 Packing Group III

#### IATA

UN-No UN3082  
 Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
 Hazard Class 9  
 Packing Group III

#### IMDG/IMO

UN-No UN3082  
 Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
 Hazard Class 9  
 Packing Group III

### 15. Regulatory information

#### United States of America Inventory

| Component                    | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | X    | ACTIVE                                        | -                           |
| 4-tert-Butyl catechol        | 98-29-3   | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

#### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                    | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | -   | X   | 214-028-8 | X     | X    | X    | X    | -     | KE-05656 |

|                       |         |   |   |           |   |   |   |   |   |          |
|-----------------------|---------|---|---|-----------|---|---|---|---|---|----------|
| 4-tert-Butyl catechol | 98-29-3 | X | - | 202-653-9 | X | X | X | X | X | KE-11368 |
|-----------------------|---------|---|---|-----------|---|---|---|---|---|----------|

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

### **U.S. State Right-to-Know Regulations**

| Component             | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------|---------------|------------|--------------|----------|--------------|
| 4-tert-Butyl catechol | X             | -          | X            | -        | -            |

### **U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

### **Authorisation/Restrictions according to EU REACH**

| Component                    | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | -                                                                   | -                                                                             | -                                                                                                     |
| 4-tert-Butyl catechol        | 98-29-3   | -                                                                   | Use restricted. See item 75.<br>(see link for restriction)                    | -                                                                                                     |

|  |  |  |          |  |
|--|--|--|----------|--|
|  |  |  | details) |  |
|--|--|--|----------|--|

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                    | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| 4-tert-Butyl catechol        | 98-29-3   | Listed         | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                    | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Benzene, 1-chloro-4-ethenyl- | 1073-67-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| 4-tert-Butyl catechol        | 98-29-3   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

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**Creation Date**

15-Dec-2011

**Revision Date**

25-Mar-2024

**Print Date**

25-Mar-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**